

		giving $DX = 0.150 \text{ cm}$ Pythagoras theorem gives $L = \sqrt{8.458^2 - 7.0^2} = 4.74 \text{ cm}$
23	D	In the setup, $VPQ = 3 / (3 + 1.5 + 22) 12.0 = 1.31 \text{ V}$. Hence, if the test cell is in range of mV, V_{pq} need to be further reduced, thus, increase $22 \text{ } \Omega$ to $1 \text{ k}\Omega$.
24	A	Closer magnetic flux lines means a stronger field to the left